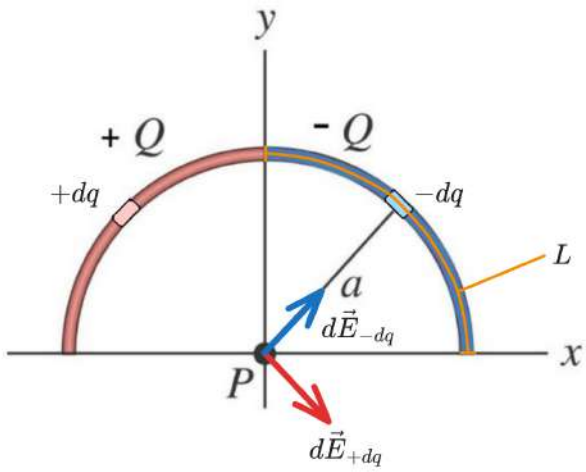
Ronnie Delos Santos, PHY 205 B

Positive charge (+Q) is uniformly distributed on the left-half of a semicircle of radius a and negative charge (-Q) is distributed on the right-half, as shown below.



$|+Q| = |-Q| = Q$

Objective 4:

1. In terms of the given variables, what is the electric field at the center of curvature (point P)?

CP

We define infinitesimal charges $+dq$ and $-dq$ which exert an electric field $d\vec{E}_{+dq}$ and $d\vec{E}_{-dq}$ respectively at point P. Due to symmetry, the vertical components of these field vectors cancel out. At the same time, the horizontal components sum together in the $+\hat{i}$ direction. Since each half contributes an identical magnitude of electric field to the horizontal field, we can integrate over just one half and multiply by 2

AP

The diagram shows a charge distribution across a semicircle with presumably no thickness. So, the distribution is said to be linear. Thus, we have a linear charge density $\lambda = \frac{Q}{L}$ where $L = \frac{\pi a}{2}$ which is the length of a quarter of the perimeter of a circle.

Next, we define an infinitesimal arc segment $ds = a\ d\theta$ across the length L. From the units of λ, which are $\frac{\mathrm{C}}{m}$, the tiny amount of charge (dq) in that segment must be the product of the density and the length of the segment to result in units of Coulombs (our tiny charge).

$$
\begin{aligned}
dq &= \lambda \left(\frac{\mathrm{C}}{\mathrm{m}}\right) \times ds(\mathrm{m}) = \lambda\ ds \\
&= \lambda(a\ d\theta) \\
&= \frac{Q}{L} a\ d\theta \\
&= \frac{Q}{\frac{\pi a}{2}} a\ d\theta \\
&= \frac{2Q}{\pi}\ d\theta
\end{aligned}
$$

AM

Since the vertical components cancel, we are only left with the horizontal components:

$$
\begin{aligned}
d\vec{E} &= dE_{+dq}\cos\theta\,\hat{i} + dE_{-dq}\cos\theta\,\hat{i} \\
dE_{dq} &= dE_{+dq} = dE_{-dq} = \frac{1}{4\pi\epsilon_0}\frac{dq}{a^2} \\
d\vec{E} &= 2\,dE_{dq}\cos\theta\,\hat{i} \\
&= 2\left(\frac{1}{4\pi\epsilon_0}\frac{dq}{a^2}\right)\cos\theta\,\hat{i} \\
&= 2\left(\frac{1}{4\pi\epsilon_0}\frac{\frac{2Q}{\pi}\,d\theta}{a^2}\right)\cos\theta\,\hat{i} \\
&= \left(\frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\right)\cos\theta\,d\theta
\end{aligned}
$$

Finally, integrating it, we get the electric field at point P:

$$
\begin{aligned}
\vec{E} &= \int_0^{\frac{\pi}{2}} d\vec{E} \\
&= \int_0^{\frac{\pi}{2}} \left(\frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\right)\cos\theta\,d\theta \\
&= \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\int_0^{\frac{\pi}{2}}\cos\theta\,d\theta \\
&= \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\cdot\sin\theta\Big|_0^{\frac{\pi}{2}} \\
&= \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\left(\sin\left(\frac{\pi}{2}\right) - \sin 0\right) \\
&= \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}(1-0) \\
&= \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}\,\frac{N}{C}
\end{aligned}
$$

Objective 1:

3. What force would this electric field exert on a charge q that was located at point P?

CP

From above, we have determined the net electric field $\vec{E}$ at P caused by the semicircular charge distribution. When a charge q is placed at P, it experiences a force $\vec{F}$ that is proportional to the both the magnitude of the charge q and the strength of the electric field at P. Since our field points in the $+\hat{i}$ direction, a positive charge q will experience a force in that same direction. Alternatively, if q was a negative charge, it would experience a force in the opposite direction.

AP

The relationship between electric force and electric field is defined as

$$\vec{F} = q\vec{E}$$

AM

Using the result from above, we obtain

$$\begin{aligned}\vec{F} &= q\frac{Q}{\pi^2\epsilon_0 a^2}\hat{i} \\ &= \frac{qQ}{\pi^2\epsilon_0 a^2}\hat{i}\ \mathrm{N}\end{aligned}$$

Objective 5:

2. Do the limiting cases of the results for Objective 4 Question 1 make sense?

CP

The limiting cases describe scenarios where $a \to 0$ (near-field) and $a \to \infty$ (far-field).

As $a \to 0$, the charge Q on each half of the semicircle concentrates towards the center P. Although the net charge is zero, the horizontal components of the field sum together causing the field magnitude to approach infinity.

As $a \to \infty$, the distance between P and the source charges become vast. Since the semicircle is a finite distribution of charge Q, the field should weaken at rate consistent with inverse-square law.

AP

From above, we have the net electric field result

$$\vec{E} = \frac{Q}{\pi^2\epsilon_0 a^2}\hat{i}$$

The presence of a^2 in the denominator ensures that the math of the field behaves according to the inverse-square law.

AM

Checking the limits:

$$\lim_{a\to 0}\frac{Q}{\pi^2\epsilon_0 a^2} = \infty$$

$$\lim_{a\to \infty}\frac{Q}{\pi^2\epsilon_0 a^2} = 0$$